

Genetic diversity, structure and conservation of the endangered Cantabrian Capercaillie in a unique peripheral habitat

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Abstract Populations at the rear edge of the species' range are often at a high risk of extinction due to their isolation, fragmentation and small population sizes. However, these populations also play a relevant role in the conservation of biodiversity since they may represent a valuable genetic resource. The endangered Cantabrian Capercaillie (*Tetrao urogallus cantabricus*) inhabits deciduous forests of the Cantabrian Mountains of Spain, at the southwestern limit of the species' range. Recently, nine Cantabrian Capercaillie leks were discovered in Pyrenean oak forests of the southern slope of the Cantabrian range, where the subspecies historically occurred. To elucidate if the origin of this peripheral

population nucleus is the result of a historical range contraction or a recent re-colonization from the core population, we sampled moulted feathers from all the known leks in the southern peripheral forests and from the adjacent main core population, based on nine microsatellite loci genotypes. No significant genetic differentiation was detected between main core and peripheral forests suggesting that gene flow is not interrupted between these nuclei. Contrary to expected, peripheral forests did not represent sink populations, since gene flow mainly occurred from southern peripheral to northern main core forests. Therefore, the origin of these birds inhabiting the peripheral nucleus seems not to be a recent colonization but relicts from the former distribution range that have remained unnoticed in a drier and warmer environment than described so far for the species. Cantabrian Capercaillie faces a high risk of extinction in the southernmost forests of its distribution, not only because of its peripheral location but also due to its small population size, low genetic diversity and low incoming gene flow. According to our results, this peripheral nucleus could represent an expanding edge for the population if Pyrenean oak forests continue to spread out southwards and consequently stress the need for conservation programs to preserve habitat availability and forest connectivity.

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Introduction

Peripheral populations can be genetically impoverished and differentiated due to isolation, low gene flow and small population sizes, thus they are frequently more sensitive to population decline than populations in the core of the species'

range (Brown 1984; Lawton 1993; Hampe and Petit 2005). In grouse, southern peripheral areas are the main arenas of peripatric speciation, since they acted as refugia during past climatic oscillations (Drovetski 2003). Additionally, these areas may encompass high environmental and genetic variation, being particularly important for maximizing adaptive diversity (Thomassen et al. 2010; Alda et al. 2013); therefore, peripheral populations are highly important for biodiversity conservation (Lesica and Allendorf 1995; Ohlemüller et al. 2008; Pearce-Higgins et al. 2011).

Capercaillie *Tetrao urogallus* is the largest grouse species. It is adapted to cold and temperate climates, inhabiting rich and continuous forests (Storch 2001; Suter et al. 2002). Its southernmost distribution limits are the Iberian and Balkan Peninsulas, both of which currently hold a differentiated evolutionary lineage from the northern and eastern Capercaillie populations, albeit secondary contact with the northern lineage has also been reported (Duriez et al. 2007; Bajc et al. 2011). The southern Capercaillie lineage is considered an evolutionary significant unit, which must be managed locally (Duriez et al. 2007; Rodríguez-Muñoz et al. 2007; Segelbacher and Pieltney 2007; Bajc et al. 2011). The Cantabrian Capercaillie *T. urogallus cantabricus* is the most isolated subspecies, occurring in the northwestern Iberian Peninsula. It is located 300 km west of the closest Capercaillie population in the Pyrenees and differs both ecologically and genetically from its conspecifics (Duriez et al. 2007; Rodríguez-Muñoz et al. 2007). In contrast to other populations inhabiting coniferous forests, the Cantabrian Capercaillie lives and feeds in deciduous forests dominated by beech *Fagus sylvatica*, birch *Betula pubescens* and oak *Quercus* sp. (Quevedo et al. 2006b; González et al. 2010, 2012). Genetically, the Cantabrian Capercaillie is the only population purely constituted by individuals from the southern evolutionary lineage and represents one of the lowest genetic diversities (Duriez et al. 2007; Rodríguez-Muñoz et al. 2007; Alda et al. 2011).

The isolated and endangered Cantabrian Capercaillie population, likely consisting of less than 500 birds, has declined by approximately 50 % in the last 20–30 years, mainly due to habitat destruction, habitat fragmentation and human disturbances (Obeso and Bañuelos 2003; Quevedo et al. 2006a; Storch et al. 2006). Currently, only 23 % of the forest cover provides suitable habitat for the Capercaillie (Quevedo et al. 2006a; Storch et al. 2006). For example, the supramediterranean forests of Pyrenean oak *Quercus pyrenaica*, which were inhabited by the Cantabrian Capercaillie until the seventeenth century, were heavily deforested in the eighteenth and nineteenth centuries, and Capercaillie was thought to have become extinct in this area (see Fig. 1a; Castroviejo 1975; Martínez 1993; García et al. 2005). However, this habitat is now naturally expanding

southward due to rural abandonment (Morán-Ordoñez et al. 2011), and nine Cantabrian Capercaillie leks have recently been found within it (González et al. 2010). The origin of these birds is unknown, but we hypothesize that these nine leks are either (1) the result of a recent re-colonization from the northern core population or (2) relicts from the historical distribution range and have remained unnoticed. In other words, this region could represent either an expanding or a contracting edge of the Cantabrian Capercaillie range.

Peripheral areas and populations are typically considered sinks, whereas core populations are usually considered sources (Dias 1996). Therefore, we hypothesize that forests from the main core habitat could act as a source for the southern peripheral forests, which may function as a sink, persisting via immigrant birds which disperse from the core habitat (Segelbacher et al. 2003). In this scenario, we would expect to see evidence of recent migration from the main core to the periphery; however, if these leks are the result of historical range contraction, gene flow could occur in both directions and at different levels, depending on their degree of isolation. For both scenarios, low levels of genetic diversity would be expected, due to either a founder effect or to long-term inbreeding. So far, all genetic studies have focused on Cantabrian Capercaillie inhabiting a main core area consisting of “typical” Eurosiberian habitats (Duriez et al. 2007; Rodríguez-Muñoz et al. 2007; Alda et al. 2011; Vázquez et al. 2012). Therefore, the study of Capercaillie living in such unique peripheral habitat is interesting because it suggests greater Capercaillie adaptability than previously thought (González et al. 2010; 2012) and is important to understand for further management and conservation of the Cantabrian Capercaillie genetic and ecological diversity. Moreover, birds from these southern peripheral forests are being used for a captive breeding program.

In our study, we aim to clarify the role of the peripheral habitat for the Cantabrian Capercaillie population (i.e. determine if it is an expanding or a contracting edge). To do this, we perform genetic analyses using microsatellite markers from samples collected from the species at all known leks from the newly described peripheral habitat and from leks in the adjacent core habitat.

Materials and methods

Study area and non-invasive sampling

The study area is located on the southern slope of the Cantabrian Mountains, in León province in northwestern Spain (centered at 42°39' N; Fig. 1a). It encompasses ~2,500 km² spanning over both sides of the putative line separating the Eurosiberian and Mediterranean biogeographical regions (González et al. 2010; Fig. 1b). The northern half of

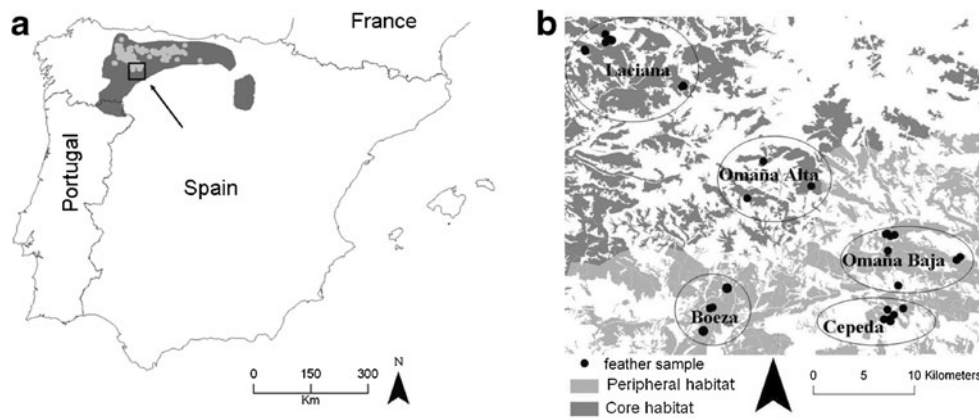


Fig. 1 **a** Map of the Iberian Peninsula indicating the Cantabrian Capercaillie historical distribution in the seventeenth century (dark grey); and the distribution in 2005 (light grey) (according to Robles et al. 2006). Arrow indicates the location of the study area. **b** Map of

the study area and sampling localities; black dots represent collection sites; light grey shading represents peripheral forests and dark grey forests from the main core population

the area is in the Eurosiberian region within the core of the Cantabrian Capercaillie distribution, where the subspecies inhabits birch and sessile oak forests in a humid and temperate climate, situated at 1,500 to 1,700 m (above sea level) a.s.l. (Obeso and Bañuelos 2003). The southern half of the area lies within the Mediterranean region with supramediterranean climate, which has two dry months in the summer. The landscape is mountainous (elevation ranges from 800 to 1,700 m a.s.l.) and dominated by Pyrenean oak forests intermingled with Scots pine *Pinus sylvestris* plantations. The entire and recently discovered peripheral Capercaillie nucleus occurs within these forests (Fig. 1b). The closest peripheral and main core leks are at least 10 km apart and in different watersheds, therefore forests show fairly different climate and ecological characteristics (González et al. 2010, 2012).

To limit the disturbance to the rare Cantabrian Capercaillie, we collected 67 freshly moulted feathers in August–September 2009 from five watersheds in the Cantabrian Mountains. Each watershed was considered as a different locality. Three watersheds were in the peripheral habitat: Omaña Baja ($n=18$), Cepeda ($n=19$) and Boeza ($n=5$); and two in the main core habitat: Laciana ($n=19$) and Omaña Alta ($n=6$). In total, we sampled in all nine leks known to occur in the peripheral forests (González et al. 2010) and in nine leks from the main core. Feather searches were conducted via a systematic zigzag walk covering 1 km² of forest surrounding each known lek, once per month. We took geographic coordinates for each sample using a global positioning system (Fig. 1b). We attributed feathers collected in the field to their respective sex according to the colour pattern; these feathers were then stored in paper envelopes until DNA extraction.

DNA extraction and microsatellite genotyping

We treated each feather as an individual sample for DNA extraction. Total genomic DNA was extracted from a $\approx$ 1-cm

segment at the base of each feather cut into small pieces using DNeasy Tissue Kit (QIAGEN, Crawley, UK) according to the manufacturer's manual with modifications for this species (Segelbacher 2002). Digestion was performed overnight in a shaking water bath at 55 °C. The buffer volume was adjusted to the size of the feather tip (400–600 μ L for large feathers, 200 μ L for small feathers). DNA was eluted in 60–100 μ L of the buffer and stored at -20 °C.

All samples were genotyped for 16 microsatellite loci: TUD1, TUD2, TUD3, TUD4, TUD5, TUD8, TUT1, TUT2, TUT3, TUT4, LLSD2, LLSD3, LLSD1, LLSD4, LLSD6, and LLSD10 (Piertney and Dallas 1997; Segelbacher et al. 2000). Microsatellites were co-amplified in five multiplex PCRs (MP1: TUD4, TUD5, TUD8 and TUT4; MP2: TUD3, TUT2 and TUT3; MP3: TUD1 and TUT1; MP4: TUD2, LLSD2 and LLSD3; MP5: LLSD1, LLSD4, LLSD6 and LLSD10). Amplifications were performed on an MJR DYAD PTC220 DNA Thermal Cycler (MJ Research, Inc., MA, USA) following QIAGEN Multiplex PCR kit protocol and using two touchdown reactions. MP1, 2 and 3 were performed decreasing the annealing temperature by 0.5 °C every second cycle for 10 cycles starting at 55 °C, followed by 30 cycles at 50 °C. MP4 and 5 were carried out decreasing the annealing temperature by 0.5 °C every second cycle for eight cycles starting at 56 °C, followed by 30 cycles at 52 °C. The total reaction volume was 10 μ L, including 5 μ L of the QIAGEN PCR Master Mix, 1 μ L of primer mix, 2 μ L of DNA, and 2 μ L of RNase-free H₂O. Reactions were performed following the M13-tailed primer method, which uses three primers for each locus: a sequence-specific forward primer with M-13 tail at its 5' end, a sequence-specific reverse primer, and the universal fluorescent-labelled M13 primer (Shuelke 2000). Fluorescently labelled PCR products were analysed on an ABI3130xl DNA analyser (Applied Biosystems, Foster City, CA, USA) and alleles were scored

using GeneMapper 4.0 software (Applied Biosystems). All samples ($n=67$) were initially screened twice with MP1 to evaluate the quality of the nuclear DNA. Samples that showed reliable amplifications and matching genotypes ($n=64$) were selected to continue the genotyping process. We conducted a pilot study using ten randomly selected samples, which were amplified four times for the 16 loci to estimate genotyping error rates using Gimlet 1.3.3 (Valière 2002). Consensus genotypes were constructed based on the following rules: for a sample to be heterozygous at a locus we had to observe both alleles twice, and for a sample to be homozygous, we had to observe the allele three times. Monomorphic loci and those showing a high rate of allelic dropout were not included in any of the following analyses (see “Results” below).

We calculated the probability of identity (PI; Taberlet and Luikart 1999) and the probability of identity among siblings (PIsib; Waits et al. 2001) for the complete set of markers using GenAlex 6 (Peakall and Smouse 2006). Also, the probability of recapture per individual was investigated through the use of regrouping function on Gimlet 1.3.3 to identify unique individuals. Identical genotypes were removed from further analyses as belonging to the same individual. Finally, we calculated the number of pairwise allele mismatches between unique individuals using GenoDive 2.0b17 (Meirmans and Van Tienderen 2004).

Data analysis

We estimated genetic diversity parameters: number of alleles (N_A), effective number of alleles (N_e), observed and expected heterozygosity (H_o and H_e), and inbreeding coefficient (F_{IS}), for all Cantabrian Capercaillie samples and for the main core and peripheral forests separately using GenoDive 2.0b17. Departures from Hardy–Weinberg equilibrium and pairwise tests for linkage disequilibrium were assessed by applying exact tests, using 100 batches and 1,000 iterations per batch, in GENEPOP 3.4 (Raymond and Rousset 1995).

We employed a Bayesian clustering method to investigate the genetic structure and spatial location of genetic discontinuities. For this purpose, we used STRUCTURE 2.3.3 (Pritchard et al. 2000) under the admixture model and correlated allele frequencies (Falush et al. 2003). We repeated the analyses with and without the LOCPRIOR option, which is recommended in those situations where the amount of available data is limited or when the standard model does not provide a clear signal of structure. We must consider that the LOCPRIOR model is substantially different from the “Model with prior population information” (Pritchard et al. 2000), since it does not place such a strong prior on the sampling locations, and consequently it does not tend to find structure when none is present, and is able to

ignore the sampling information when the ancestry of individuals is uncorrelated with sampling locations (Hubisz et al. 2009). We performed ten independent runs for each K value from $K=1$ to $K=8$. Each run had 5×10^5 iterations with a burn-in of 1×10^5 iterations. Mean log probabilities were used to calculate ΔK , and find the K value with the highest probability (Evanno et al. 2005). Once K was estimated, we ran five independent analyses with K fixed at the highest probability value for 5×10^6 iterations with a burn-in of 1×10^6 iterations value. The final Q coefficients were obtained by averaging these five runs using CLUMPP 1.1.2 and the ‘greedy’ algorithm and the ‘all possible input order’ options (Jakobsson and Rosenberg 2007).

To test if there is a significant genetic differentiation between Cantabrian Capercaillie samples in the main core and peripheral forests, or if genetic differentiation is best explained by their sampling localities, we performed an analysis of molecular variance (AMOVA) in GenoDive 2.0b17. Also, we explored the occurrence of an isolation-by-distance pattern of spatial genetic structure relationship. We calculated Euclidean distances between individuals and tested their correlation with their genetic distance (Smouse and Peakall 1999) using Mantel tests (Mantel 1967). To find out whether patterns of population differentiation are only due to isolation by distance or also due to a putative barrier between core and peripheral forests, we performed a partial Mantel test (Smouse et al. 1986). In this test, the association between geographic and genetic distances was tested while controlling for the influence of forest type (core or peripheral) using a binary matrix to code for individuals located in the same (0) or in different forests (1). These analyses were performed in GenoDive 2.0b17 and their statistical significance was assessed by 1×10^5 randomizations.

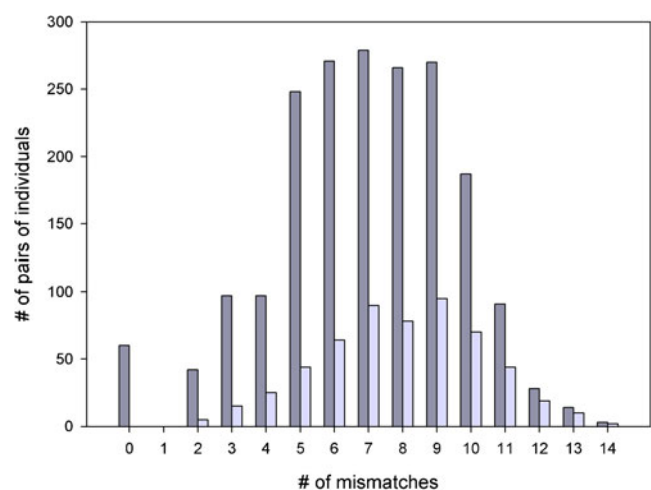


Fig. 2 Number of mismatching markers between pairs of genotypes, considering nine polymorphic microsatellites. Dark grey bars represent 64 samples analysed and light grey bars represent 34 unique genotypes

Recent migration among localities and between the main core and peripheral forests was investigated using BAYESASS 1.3 (Wilson and Rannala 2003). To ensure convergence of the MCMC, we performed ten runs of 3×10^6 iterations including a burn-in of 1×10^6 iterations and a sampling frequency of 2,000. Delta values were tuned individually to obtain acceptance rates within 40–60 % of the total.

Results

Overall, 64 Cantabrian Capercaillie feathers were genotyped for all 16 microsatellites identifying 34 unique genotypes: 17 in peripheral (Omaña Baja, $n=8$; Cepeda, $n=5$; Boeza, $n=4$) and 17 in main core forests (Laciana, $n=13$, Omaña Alta, $n=4$). These genotypes corresponded to 15 females (seven in the periphery and eight in the main core) and 19 males (ten in the periphery and nine in the main core). Thirteen individuals were recaptured, ten of which were observed two or three times in the sample and three were recaptured five, six and seven times each. All recaptures were found within the same localities and considered to represent the same individual. The mean number of pairwise allelic mismatches between individuals was 7.913 (SD, 2.341) (Fig. 2).

According to our data, the expected number of individuals with the same genotype in the population was 6.079×10^{-4} ($PI=1.788 \times 10^{-5}$ and $PI_{Sib}=6.496 \times 10^{-3}$). The only observed false allele rate for any locus was 7.5 % for LLSD2 and the allelic dropout rate per locus was 0 % for 11 loci and above 0 % for TUT8 (6.3 %), LLSD3 (8.3 %), and TUT3 (25 %). Because of the moderate likelihood of allelic dropout at locus TUT3, we re-genotyped a random sample of homozygotes for TUT3. Although we did not find any cases of dropout, we decided to remove this locus from subsequent analyses. The maximum error rate per individual calculated for the ten individuals genotyped four times, and not considering locus TUT3, was 0 % for seven individuals, and 12.5 %, 20 %, and 25 % for the remaining three individuals.

No evidence for linkage disequilibrium between loci was found (all P values >0.106). Genetic diversity for the Cantabrian samples was low (Table 1). Overall, average allele number was $N_A=3.444$ (SD, 1.589) and observed heterozygosity $H_o=0.348$ (SD, 0.198). Six loci were invariable: TUT2, TUT4, LLSD1, LLSD2, LLSD4 and LLSD10. Additionally, in the peripheral forest localities loci LLSD6 and TUT4 were also monomorphic (Table 1).

The STRUCTURE analyses identified the most likely genetic structure as $K=3$, either using LOCPRIOR option or not (Fig. 3). However, when the analysis for the Cantabrian

Table 1 Genetic diversity of capercaillie based on polymorphic microsatellite loci

Nucleus	Locus (bp)	TUD1 (242–250)	TUD2 (196–206)	TUD3 (210–214)	TUD4 (80–110)	TUD5 (109–121)	TUD8 (123–129)	TUT1 (229–245)	LLSD3 (162–166)	LLSD6 (118–124)	Overall
All	$n=34$										
	N_A	3	3	3	2	7	3	5	2	3	3.444
	N_e	1.622	1.734	1.826	1.066	2.028	1.931	1.99	1.5	1.058	1.639
	H_o	0.417	0.442	0.336	0.015	0.453	0.608	0.442	0.387	0.031	0.348
	H_e	0.604	0.506	0.516	0.076	0.628	0.525	0.734	0.399	0.062	0.45
	F_{IS}	−0.044	0.102	0.3	0.785	0.26	−0.271	0.271	−0.01	0.497	0.144
Peripheral	$n=17$										
	N_A	3	3	3	1	3	3	5	2	1	2.667
	N_e	1.46	1.753	1.738	1	1.589	2.024	1.567	1.485	1	1.513
	H_o	0.208	0.442	0.483	0	0.325	0.667	0.333	0.35	0	0.312
	H_e	0.556	0.535	0.493	0	0.489	0.545	0.667	0.432	0	0.413
	F_{IS}	0.267	0.088	−0.166	–	0.241	−0.324	0.258	0.158	–	0.062
Main Core	$n=17$										
	N_A	3	3	2	2	7	2	5	2	3	3.222
	N_e	1.946	1.706	1.977	1.184	3.463	1.807	3.338	1.522	1.158	2.011
	H_o	0.731	0.442	0.115	0.038	0.644	0.519	0.606	0.442	0.077	0.402
	H_e	0.659	0.481	0.54	0.187	0.797	0.492	0.812	0.356	0.155	0.498
	F_{IS}	−0.315	0.114	0.676	0.778	0.265	−0.206	0.274	−0.205	0.484	0.199

Bold values indicate significant departures from Hardy–Weinberg equilibrium ($P<0.05$) n number of samples, N_A number of alleles, N_e effective number of alleles, H_o observed heterozygosity, H_e expected heterozygosity, F_{IS} inbreeding index

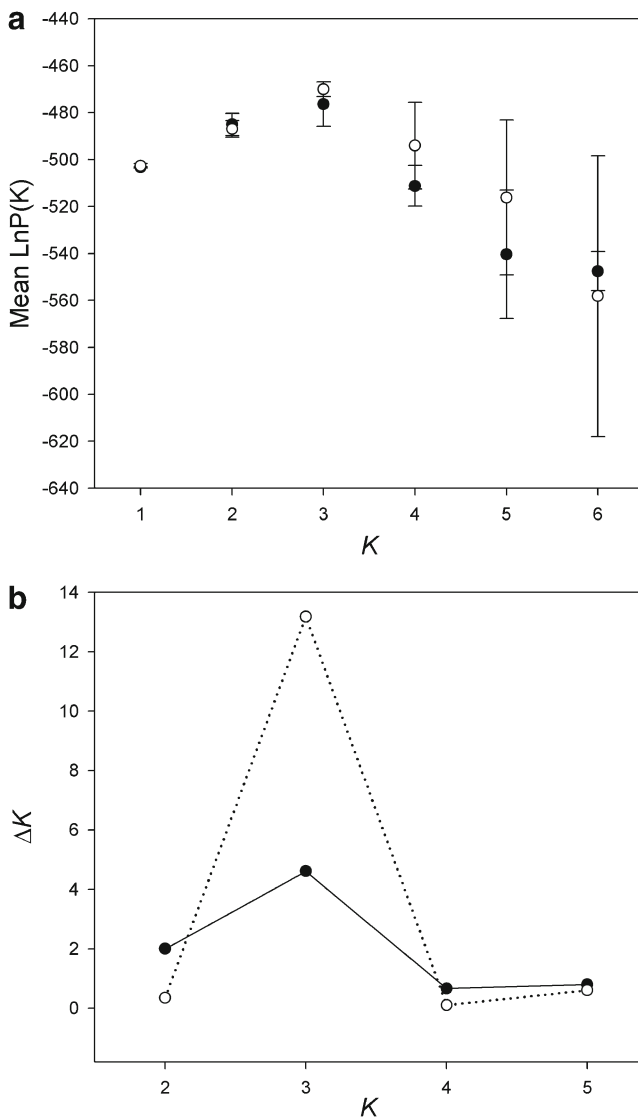


Fig. 3 Plots representing mean log probabilities $\text{LnP}(K)$ and associated standard errors (a) and the second derivative of the mean log probability (ΔK) (Evanno et al. 2005) (b) obtained in the STRUcTURE analyses with (open circles) and without (solid circles) the LOCPRIOR option

samples was performed without the LOCPRIOR option, individuals were assigned to the clusters with a lower probability (q), therefore we will be showing the results obtained from the LOCPRIOR model. Cepeda and Boeza were the most differentiated localities and assigned with a high probability to clusters 1 and 2, respectively. Laciana had individuals with a high proportion of inferred ancestry from clusters 1 and 3, whereas Omaña Alta and Omaña Baja consisted of individuals with intermediate membership proportions of all three genetic groups (Fig. 4).

The AMOVA indicated a significant genetic differentiation among all Cantabrian Capercaillie localities ($F_{ST}=0.111$, $P<0.001$), although most of the genetic variation was explained among individuals ($F_{IS}=0.144$, $P=0.006$).

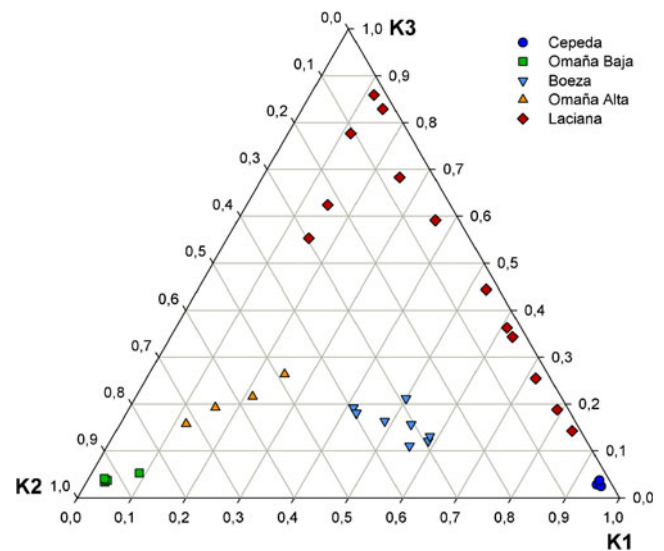


Fig. 4 Triangular plot representing individual assignment probabilities for each of the genetic groups inferred in STRUcTURE ($K=3$). Each triangle corner represents an assignment probability (q) of 1.0 for each of the inferred genetic groups

On the other hand, no significant genetic differentiation was observed between forests in the main core and the periphery of the population, suggesting that gene flow is not interrupted between these nuclei ($F_{CT}=-0.065$, $P=0.303$). Current migration rates, as inferred with BAYESASS, were nonsignificant as all 95 % confidence intervals overlapped with those simulated when there is no information in the data (Table 2). Failure to detect recent dispersal events between populations despite high rates of gene flow may have resulted from small sample sizes (Galbusera et al. 2000). Still, some estimates were significantly different from the others and indicated a clear tendency for gene flow to primarily occur from the periphery into the main core forests ($m=0.27$; 95 % CI, 0.22–0.33; Table 2). Also, the southern locality of Cepeda showed the highest migration rate to the northernmost locality of Laciana ($m=0.16$; 95 % CI, 2.57×10^{-5} –0.54; Table 2).

Comparisons of pairwise genetic (Smouse and Peakall 1999) and Euclidean geographical distances between all individuals revealed a significant isolation-by-distance pattern of genetic structure (Mantel's $r=0.111$, one-sided $P=0.009$). When we considered not only geographic distance between individuals but also the effect of a putative barrier between core and peripheral forest, we found a higher significant correlation between genetic distance and geographic distance (Partial Mantel's $r=0.175$, one-sided $P<0.0001$). Additionally, spatial genetic structure differed between forests in the main core and in the periphery when these were considered separately, as a significant isolation-by-distance relationship was observed for Capercaillie individuals in the periphery (Mantel's $r=0.476$, one-sided

Table 2 Migration rates inferred using BAYESASS between habitats and localities of Cantabrian capercaillie

		From				
Into	Peripheral	Peripheral	Main core			
		0.92 (0.73, 0.99)	0.08 (1.75×10^{-3} , 0.26)			
	Main core	0.27 (0.12, 0.33)	0.72 (0.67, 0.87)			
Simulated 95 % confidence interval when there is no information in the data: for the non-migration rates (0.67–0.99) and migration rate between two populations (0.01–0.32).						
		From				
Into	Cepeda	Cepeda	Omaña Baja	Omaña Alta	Boeza	Laciana
		0.85 (0.67, 0.99)	0.02 (9.57×10^{-6} , 0.09)	0.03 (5.61×10^{-6} , 0.22)	0.02 (5.8×10^{-6} , 0.11)	0.08 (2.94×10^{-6} , 0.29)
	Omaña Baja	0.02 (2.15×10^{-6} , 0.11)	0.91 (0.69, 0.99)	0.02 (2.68×10^{-6} , 0.22)	0.03 (2.27×10^{-6} , 0.22)	0.02 (1.99×10^{-6} , 0.11)
	Omaña Alta	0.09 (2.43×10^{-5} , 0.28)	0.02 (5.30×10^{-5} , 0.12)	0.78 (0.67, 0.99)	0.03 (2.37×10^{-5} , 0.14)	0.08 (2.06×10^{-5} , 0.28)
	Boeza	0.03 (1.01×10^{-4} , 0.13)	0.08 (7.02×10^{-4} , 0.23)	0.11 (1.80×10^{-3} , 0.27)	0.75 (0.67, 0.91)	0.04 (1.53×10^{-4} , 0.15)
	Laciana	0.16 (2.57×10^{-4} , 0.54)	0.01 (9.57×10^{-6} , 0.09)	0.03 (5.57×10^{-5} , 0.14)	0.01 (5.88×10^{-6} , 0.11)	0.79 (0.67, 0.96)
Simulated 95 % confidence interval when there is no information in the data: for the non-migration rates (0.67–0.99) and migration rate between five populations (1.79×10^{-5} –0.18).						

Values in columns represent migration rates into localities indicated to the left. Values in italics in the diagonal represent the proportion of non-migrant individuals

$P < 0.0001$) but not for those in the main core (Mantel's $r = 0.071$, one-sided $P = 0.273$; Fig. 4).

Discussion

Habitat loss and fragmentation have a profound effect on the genetic isolation, structure, and dispersal of Capercaillie (Segelbacher and Storch 2002; Segelbacher et al. 2003, 2008; Alda et al. 2011). After the contraction of the Cantabrian Capercaillie distribution in the last centuries caused by deforestation, mainly in the southern part of the range, records of Capercaillie presence were only reported from the northern Eurosiberian forests (Madoz 1848; Castroviejo 1975; Martínez 1993; Fig. 1a). However, recently it was found that the subspecies also occurs in peripheral Pyrenean oak forests (González et al. 2010). Our results show a lack of significant genetic differentiation between forest in the main core and in the periphery of the Cantabrian Capercaillie population. Importantly, and contrary to the expectation that populations at the edge of the species' range may function as sinks (Segelbacher et al. 2003), here the northern forests seem to receive immigrants rather than acting as source habitat that supply emigrants to the southern peripheral forests (Table 2). Therefore, gene

flow into the periphery is low since the proportion of non-migrants is high (see Table 2). Furthermore, these results suggest that peripheral forests have not been recently colonized by Capercaillie from the main core of the population, rather it is likely that these individuals have remained unnoticed until now although maintaining genetic contact with the northern Cantabrian Capercaillie.

It is obvious that if the peripheral forest population is not a sink, any existing gene flow must occur to the north, as no habitat is available southwards. Thus, we would expect that the south to north gene flow could take place by two pathways: (1) Cepeda and Omaña Baja exchanging individuals with Laciana via Omaña Alta and (2) Boeza exchanging individuals with Laciana via Omaña Alta. The first route was supported by the Bayesian clustering method, which shows a high connectivity among the localities of Cepeda, Omaña Baja and Laciana (Fig. 2), the migration analysis suggested that this was primarily due to the immigration of individuals from Cepeda to the main core (Table 2). However, there is little gene exchange between Boeza and Omaña Alta-Laciana (Table 2). Because we surveyed each lek and surrounding areas known to be occupied by Capercaillie in the southern limit of its distribution (González et al. 2010), the reduced connectivity along this route could be related to: (1) the significant degradation of

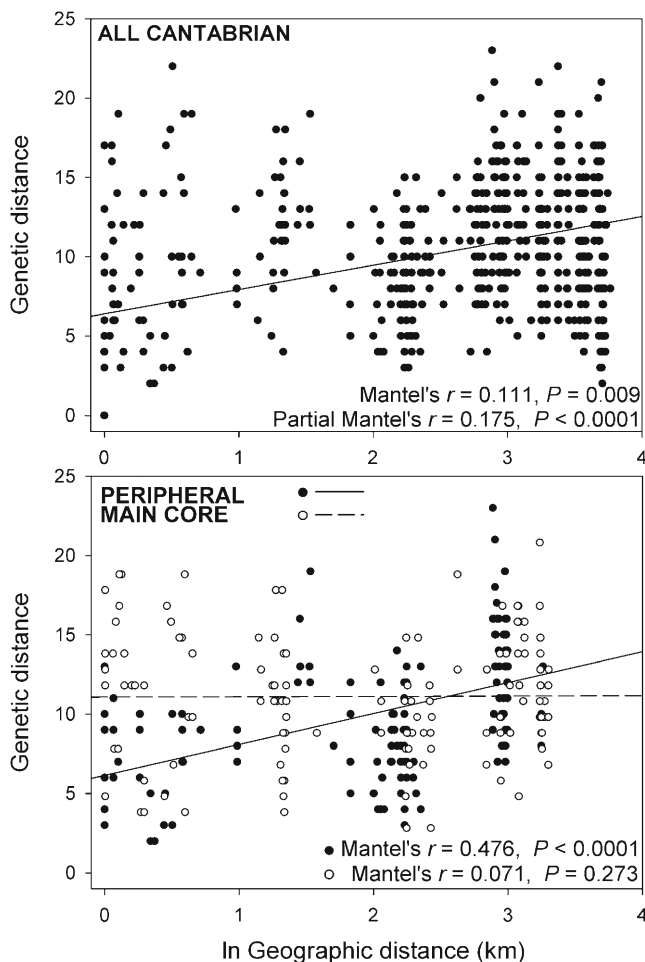


Fig. 5 Correlation plots obtained for the Mantel test between genetic (Smouse and Peakall 1999) and geographic distances for all Cantabrian individuals, and for the main core and peripheral forests separately

habitat surrounding Boeza, (2) the presence of 2,000-m a.s.l. peaks to its north and (3) to the relatively large distance between Boeza and northern localities (Omaña Alta and Laciana) that may altogether reduce the birds chances to move between Boeza and the other locations. Omaña Alta appears to act as a stepping stone, connecting main core and peripheral forests and thus playing a crucial role in the maintenance of the regional metapopulation (Storch 1997). Our data also suggest that Boeza faces the highest risk of losing Capercaillie because it is the most geographically and genetically isolated locality.

The presence of gene flow between peripheral and core forests and the non-significant AMOVA results suggest that there are no significant barriers to dispersal between population nuclei. Previous studies have found that forest quality and spatial distribution could be responsible for different patterns of genetic structure (Alda et al. 2011). Here we found an overall positive significant correlation between genetic and geographical distances. However, when looking individually at each of the population nuclei, only the

peripheral forests showed a pattern of isolation-by-distance (Fig. 5). Such patterns of isolation-by-distance have been reported for the Capercaillie at large geographical scales and are expected for species with lekking reproductive systems and low dispersal (Bouzat and Johnson 2004; Segelbacher et al. 2007), but at smaller scales, this pattern is not always observed (Segelbacher and Storch 2002; Segelbacher et al. 2003; Alda et al. 2011; Vázquez et al. 2012). Although these differences might be an effect of sampling scale, as this has proven to be highly important for analyses of dispersal (Segelbacher et al. 2008; Mäki-Petäys et al. 2007), in our case, sampling and geographical areas (although small) were balanced between habitats. Thus, it seems that the overall observed genetic structure could be mainly due to individuals in the periphery. This is supported by greater genetic differentiation between localities in the southern periphery than between those in the main core, as revealed by the Bayesian clustering method (Fig. 4). The more fragmented peripheral forests may limit dispersal to short distances and avoid dispersal to the deforested south, and thus, resulting in the greater genetic differentiation among localities than in the core population. Also, fragmented or low habitat availability in peripheral forests may increase the probabilities that males establish on leks close to their natal sites, thus increasing kinship between individuals (Regnaut et al. 2006) and leading to within-site high genetic homogeneity (e.g. in Boeza and Cepeda) and genetic differentiation between sites. In turn, these habitat-related factors may be limiting the capacity of the peripheral forests to hold large bird numbers while forcing Capercaillie dispersal to the north.

Capercaillie inhabiting southern peripheral forests face additional threats for its survival, such as low genetic diversity, low number of birds, and very importantly, low incoming gene flow from other regions, which is a key process for population persistence (Hanski and Gilpin 1991; Segelbacher et al. 2003). Genetic diversity in the periphery ($H_o=0.348$, $H_e=0.450$) is lower than that previously reported for individuals from the Cantabrian core population, regardless of our lower sampling size ($H_o=0.440$, $H_e=0.568$ and $H_o=0.45$, $H_e=0.59$ in Alda et al. 2011 and Vázquez et al. 2012, respectively), and lower than the values obtained for other small and highly endangered populations ($H_o=0.46$, $H_e=0.58$ in Poland, Rutkowski et al. 2005; and $H_o=0.53$, $H_e=0.57$ in the Jura Mountains in Switzerland, Regnaut 2004; Regnaut et al. 2006). In turn, this implies additional risk for extinction, since reduced genetic diversity can lead to the expression of deleterious alleles through inbreeding depression, hindering the capacity of a population to respond to environmental changes (Westemeier et al. 1998; Frankham 2005). Furthermore, it is argued that a stochastic environment, in some cases, can be more influential in driving a population to extinction than its genetic diversity. Thus, preserving habitat and population

connectivity should be a major conservation issue (Lin and Liu 2006).

In the Cantabrian range, the forest cover is low (23 %) and only 5 % of the Cantabrian landscape may be considered suitable habitat for Capercaillie (Abajo Chic 2007). On the other hand, the Pyrenean oak forest is naturally expanding to the south and expected to continue expanding in the future (Morán-Ordoñez et al. 2011), representing a potential increase of Capercaillie habitat availability. Therefore, in the short-term, preserving habitat availability should be of greatest concern. Conservation measures should promote the preservation of any area occupied by the Capercaillie and promote connectivity both within and between core and peripheral population nuclei in order to increase the long-term survival of the Cantabrian Capercaillie. A particular effort should be made to allow the natural recovery of the Pyrenean oak forests since its predicted expansion could allow the Cantabrian Capercaillie to recolonize part of its lost historical distribution.

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